Circular motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate inward net force to a change in velocity direction.

SUCCESS CRITERIA - I can...

1. Draw tangent velocity and inward acceleration.
2. Use v^2/r or a supplied acceleration.
3. Identify the real interaction providing centripetal force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A projectile's velocity is (**9**, **-12**) m/s.
1. **Find its speed**.
2. What information would locate it?

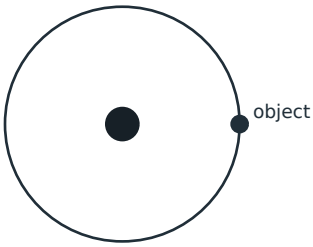
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $a_c = v^2/r$; $F_{net} = mv^2/r$.
c means centripetal: toward the center. This is the net inward force, not an extra force.

Worked reasoning: A **2 kg** object moves at **3 m/s** in a circle of radius **3 m**. Inward acceleration is v^2/r=**3 m/s²**; net force is **6 N** inward.

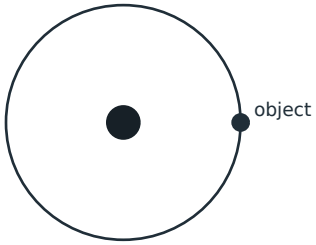
Your turn: Draw tangent velocity and inward net force at two different points.



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03 Read and explain the diagram or equation

1. **Add velocity** and acceleration arrows to the marked point on the circle.
2. **Name one interaction** that could provide the inward force.

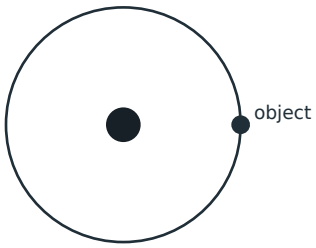


The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Apply the physics

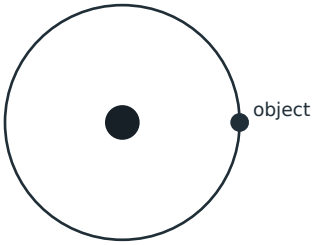
A **0.50 kg** puck on a string moves at **4 m/s** in a horizontal circle of radius **2 m**. **Draw velocity** and net-force arrows. **Calculate inward acceleration** and net force. Then **predict both quantities** at twice the speed and the same radius.



Marzano check - circle: ① ② ③

05 Explain your reasoning

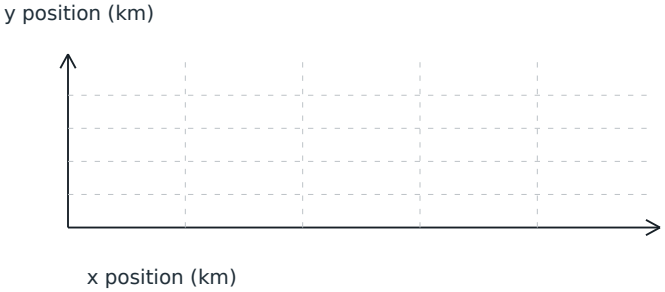
A string breaks while a ball moves in a circle. Neglect other forces over the next instant. **Sketch its initial path after** release.



Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

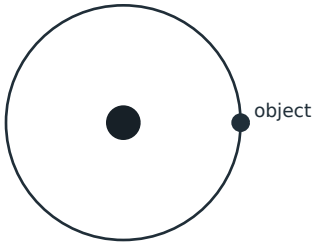
A simplified calculation predicts position (**-80,100**) km with bounds ±**8 km** in x and ±**10 km** in y. **Write a short briefing giving** both intervals and two reasons it is not a real asteroid collision forecast.
Label the given and predicted quantities on the diagram.



Marzano check - circle: 1 2 3

07 Exit check

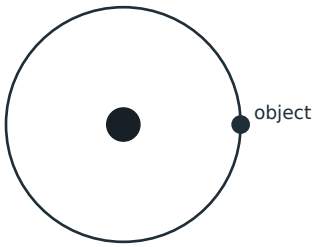
At fixed speed and mass, double circle radius. How do acceleration and net force change?
1. Explain from equations.
2. **Label the radius** and **show velocity** and inward acceleration on the circle.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

The maximum horizontal force on a **0.5 kg** object is **8 N**. For a **1 m** circle, find the maximum speed. Then double the radius and **find the new limit**. **Label the radius** and **show velocity** and inward acceleration on the circle.



Marzano check - circle: 1 2 3